

DPT Exercises

Exercise sheet 8

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Exercise 1 (heavy planes). A plane that can transport at most 100 passengers is filled with just enough jet fuel to carry up to 8 tons. For a fully booked flight, let X_i be a random variable modeling the i -th passengers weight in kg.

- (i) Write down one (convincing) argument for why the $(X_i)_{i=1}^{100}$ should be modeled as independent or not as independent. From now on, assume they are iid with $\mathbb{E}[X_i] = 75$, and $\sigma[X_i] = 20$ for all $i \in [100]$.
- (ii) Can you compute the exact probability that the maximum weight capacity of 8 tons is exceeded? Either compute it or explain why you can't?
- (iii) Use the clt to compute an approximate probability that the maximum weight capacity will be exceeded.
[Hint: You can use wolframalpha or your favorite programming language/calculator to evaluate the cdf of standard Gaussians $\Phi(\cdot)$. Instead, you can also keep the results in terms of $\Phi(\cdot)$.]
- (iv) If the profit margin per passenger is 3 Euros and the cost of going over capacity is 3000 Euros per flight (for topping up jet fuel), is the plane on average profitable for the airline (assuming it's always fully booked)?
- (v) **[Optional:]** If you were working for the airline trying to maximize profits. How would you improve the model (e.g., model how fully planes are booked, potential dependencies, varying ticket costs, etc) and what would you optimize? (This is a purely imaginative exercise, no need to do any computations. Just train your "modeling and assumptions muscles.")

Solution:

- (i) **For independence:** People booking airplane tickets are quite diverse and book for different reasons from all walks of life. There is no reason to believe that the body weight of one passenger depends on another one. Note that even if the plane is connecting specific nations that are on average above or below the global mean weight, the individual weights may still be iid from that weight distribution.

Against independence: Families or larger travel groups often book flights together. People who live together or have lots in common, also often share eating and activity habits. Hence, their body weights are likely also dependent. Having the full team of the Japan Sumo Association on board or the Austrian State Ballet makes a big difference.

- (ii) Let $S_n = \sum_{j=1}^n X_j$ be the total weight of the first n passengers. For $n = 100$, due to the iid nature of the X_i , we want to compute $P(S_{100} > 8,000)$ for which we would need to know the exact distribution of the X_i to compute it exactly. For example, the assumed distribution of body weight could be heavily left or right skewed even for the provided fixed mean and variance, which would change the probability of exceeding 8,000.

- (iii) We first compute

$$\mathbb{E}[S_{100}] = 100 \cdot \mathbb{E}[X_1] = 7,500, \quad \text{var}[S_{100}] = 100 \cdot \text{var}[X_1] = 100 \cdot 20^2 = 40,000.$$

Hence, $\sigma[S_{100}] = 200$ and, based on the clt, we suppose that S_n is normally distributed. We then compute the probability of exceeding the maximum of 8,000 kg as

$$P(S_{100} > 8,000) = P\left(\frac{S_{100} - 7,500}{200} > \frac{8,000 - 7,500}{200}\right) = 1 - \Phi\left(\frac{5}{2}\right) \approx 0.00621.$$

- (iv) With a profit margin of 3 Euros per person, we have a profit of 300 Euros for a full plane per flight. On average, we have to pay the 3000 Euros in 0.621% of flights, i.e., on average, $0.00621 \cdot 3000 = 18.63$ Euros per flight. Hence, the expected profit per flight is $300 - 18.63 > 0$. The plane is profitable for the airline.
- (v) No right or wrong answer, just discuss. Of course, this is an actual profession, see [here](#), or [here](#)

Exercise 2 (insurance claims and risk). An insurance company has $n = 10^4$ clients. For each client i , let X_i be the amount claimed in a year. Assume the claims $X_1, \dots, X_n$ are iid RVRVs with mean $\mathbb{E}[X_i] = \mu = 500$ and variance $\text{var}[X_i] = \sigma^2 = 800^2 = 640,000$. The company charges each client a premium of Z Euros.

- (i) Let $S_n = \sum_{i=1}^n X_i$ be the total amount claimed. Calculate the minimum premium Z the company needs to charge to make an expected profit of half a million per year.
- (ii) Use the weak law of large numbers to describe the behavior of the average claim per client $\bar{X}_n = S_n/n$ as n becomes very large.
- (iii) The company wants to be at least 95% sure that the total premiums collected will cover the total claims S_n . Use Chebyshev's inequality to assess if the current number of clients $n = 10^4$ is sufficient with a premium of $Z = 550$.
[Hint: The company makes a loss if $S_n > n \cdot Z$. You want to bound $P(S_n > n \cdot Z)$.]
- (iv) Now, use the clt to approximate the probability that the total claims S_n exceed the total premiums $n \cdot Z$. Compare this approximation with the bound obtained from Chebyshev's inequality. Why might the clt provide a different (and potentially more accurate) value for this probability? Which would you trust more?
[Hint: We need $P(S_n > n \cdot Z)$, which is easiest when S_n is standardized. You can use wolframalpha or your favorite programming language/calculator to evaluate the cdf of standard Gaussians.]

Solution:

- (i) The total premium collected is $n \cdot Z$. The expected total claim is $\mathbb{E}[S_n] = \mathbb{E}[\sum_{i=1}^n X_i] = \sum_{i=1}^n \mathbb{E}[X_i] = n\mu = 5 \cdot 10^6$. The total expected profit is $n \cdot Z - \mathbb{E}[S_n]$. To make a profit of half a million, we need $n \cdot Z - \mathbb{E}[S_n] = 5 \cdot 10^5$ which implies $Z = (5 \cdot 10^5 + 5 \cdot 10^6)/10^4 = 550$.
- (ii) According to the wlln, the sample mean $\bar{X}_n$ converges in probability to the true mean $\mu = 500$ as $n \rightarrow \infty$, i.e.,

$$\bar{X}_n \xrightarrow{P} \mu = 500 \quad \text{as } n \rightarrow \infty.$$

This means that for a large number of clients, the average claim per client is very likely to be close to 500.

- (iii) The company wants $P(S_n \leq n \cdot Z) \geq 0.95$, which is equivalent to $P(S_n > n \cdot Z) \leq 0.05$. We have $\mathbb{E}[S_n] = n\mu = 5 \cdot 10^6$ and $\text{var}[S_n] = n\sigma^2 = 6.4 \cdot 10^9$. The standard deviation is $\sigma_{S_n} = \sqrt{n}\sigma = 8 \cdot 10^4$. The total premium is $n \cdot Z = 5.5 \cdot 10^6$. We want to bound $P(S_n > 5.5 \cdot 10^6)$. This is equivalent to $P(S_n - \mathbb{E}[S_n] > 5 \cdot 10^5)$. This probability is less than or equal to $P(|S_n - \mathbb{E}[S_n]| \geq 5 \cdot 10^5)$. Using Chebyshev's inequality with $\epsilon = 5 \cdot 10^5$ we compute

$$P(|S_n - \mathbb{E}[S_n]| \geq 5 \cdot 10^5) \leq \frac{\text{var}[S_n]}{\epsilon^2} = \frac{6.4 \cdot 10^9}{2.5 \cdot 10^{11}} = \frac{6.4}{250} = 0.0256.$$

Since $0.0256 \leq 0.05$, Chebyshev's inequality suggests that with $n = 10000$ clients and a premium of 550, the probability of total claims exceeding total premiums is at most 2.56%, meeting the requirement.

- (iv) By the clt, S_n is approximately normally distributed for large n with $S_n \sim \mathcal{N}(\mathbb{E}[S_n], \text{var}[S_n]) = \mathcal{N}(5 \cdot 10^6, 6.4 \cdot 10^9)$. We want to calculate $P(S_n > n \cdot Z) = P(S_n > 5.5 \cdot 10^6)$. We standardize this distribution via

$$Y = \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\text{var}[S_n]}}$$

such that we are looking for

$$P(S_n > 5.5 \cdot 10^6) \approx P\left(Y > \frac{5.5 \cdot 10^6 - 5 \cdot 10^6}{8 \cdot 10^4}\right) = P(Y > 6.25) = 1 - \Phi(6.25) \approx 2 \cdot 10^{-10}.$$

where Φ is the cdf of a standard Gaussian.

Comparison: Chebyshev's inequality gave an upper bound of 0.0256, which is not a lot, but still a substantial risk. The clt asserts that the probability of a loss in any given year is essentially negligible (on the order of 1 year for the entire age of the universe). Whom should you trust more?

Chebyshev gives an upper bound. It definitely holds, no matter what the distribution of claims actually is (given the asserted means and variances are correct). Instead, the clt provides an approximation, since S_n is only normal (in distribution) for $n \rightarrow \infty$. For any finite n , albeit large, the approximation may be better or worse, depending on the specifics of the distribution of the X_i . For heavily skewed distributions, the clt approximation may substantially underestimate the risk even for $n = 10^4$. If the normal distribution approximation holds well, the clt leverages much more information and the bound will be much more precise. Whom to trust—you decide!

Exercise 3. Let X, Y RVRVs and $(X_n)_{n \in \mathbb{N}}, (Y_n)_{n \in \mathbb{N}}$ sequences of RVRVs. Verify the following statements.

- (i) If $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$ then $X_n + Y_n \xrightarrow{P} X + Y$.
[Hint: Use the triangle inequality $|(X_n - X) + (Y_n - Y)| \leq |X_n - X| + |Y_n - Y|$. Decompose the probability to be bounded by $P(A \cup B) \leq P(A) + P(B)$ (this is called a *union bound* where A is an event only about X_n, X and B is only about Y_n, Y .)]
- (ii) If $\mathbb{E}[|X_n - X|] \rightarrow 0$, then $X_n \xrightarrow{P} X$.
[Hint: Use Markov's inequality for $|X_n - X|$.]
- (iii) If $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{P} Y$ then $P(X = Y) = 1$, i.e., X and Y are equal "almost surely."
[Hint: Again use the triangle inequality $|X - Y| \leq |X - X_n| + |X_n - Y|$, decompose $P(|X - Y| \geq \epsilon)$ via a union bound (for any n), to get to $P(|X - Y| \geq \epsilon) = 0$ for any $\epsilon > 0$ and use a limiting argument to get the statement also for $\epsilon = 0$ (for example via σ -subadditivity.)]
- (iv) If $\mathbb{E}[|X_n - X|] \rightarrow 0$ and $\mathbb{E}[|X_n - Y|] \rightarrow 0$ then $P(X = Y) = 1$.

Solution:

- (i) We want to show that for any $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|(X_n + Y_n) - (X + Y)| \geq \epsilon) = 0$. Using the triangle inequality, $|(X_n - X) + (Y_n - Y)| \leq |X_n - X| + |Y_n - Y|$. If $|(X_n + Y_n) - (X + Y)| \geq \epsilon$, then it must be that $|X_n - X| + |Y_n - Y| \geq \epsilon$. This implies that either $|X_n - X| \geq \epsilon/2$ or $|Y_n - Y| \geq \epsilon/2$ (if both were less than $\epsilon/2$, their sum would be less than ϵ). So, the event $\{|(X_n + Y_n) - (X + Y)| \geq \epsilon\}$ is a subset of the event $\{|X_n - X| \geq \epsilon/2\} \cup \{|Y_n - Y| \geq \epsilon/2\}$. Therefore,

$$P(|(X_n + Y_n) - (X + Y)| \geq \epsilon) \leq P(\{|X_n - X| \geq \epsilon/2\} \cup \{|Y_n - Y| \geq \epsilon/2\}).$$

Using the union bound ($P(A \cup B) \leq P(A) + P(B)$):

$$P(|(X_n + Y_n) - (X + Y)| \geq \epsilon) \leq P(|X_n - X| \geq \epsilon/2) + P(|Y_n - Y| \geq \epsilon/2).$$

Since $X_n \xrightarrow{P} X$, $Y_n \xrightarrow{P} Y$, both terms go to zero as $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} P(|(X_n + Y_n) - (X + Y)| \geq \epsilon) \leq 0 + 0 = 0.$$

Since probabilities are non-negative, the limit must be 0. So $X_n + Y_n \xrightarrow{P} X + Y$.

- (ii) This states that convergence in expected mean absolute error implies convergence in probability. Let $\epsilon > 0$. By Markov's inequality applied to the non-negative RVRV $|X_n - X|$

$$P(|X_n - X| \geq \epsilon) \leq \frac{\mathbb{E}[|X_n - X|]}{\epsilon}.$$

Since $\mathbb{E}[|X_n - X|] \rightarrow 0$ as $n \rightarrow \infty$, for any fixed $\epsilon > 0$ we have

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{\mathbb{E}[|X_n - X|]}{\epsilon} = 0.$$

Since probability is non-negative, $\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$, i.e., $X_n \xrightarrow{P} X$.

- (iii) We want to show that $P(X \neq Y) = 0$. For any $\epsilon > 0$, consider the event $\{|X - Y| \geq \epsilon\}$. Using the triangle inequality, $|X - Y| = |X - X_n + X_n - Y| \leq |X - X_n| + |X_n - Y|$. If $|X - Y| \geq \epsilon$, then $|X - X_n| + |X_n - Y| \geq \epsilon$. This implies that either $|X - X_n| \geq \epsilon/2$ or $|X_n - Y| \geq \epsilon/2$. So, $\{|X - Y| \geq \epsilon\} \subseteq \{|X_n - X| \geq \epsilon/2\} \cup \{|X_n - Y| \geq \epsilon/2\}$. Therefore, for any n :

$$P(|X - Y| \geq \epsilon) \leq P(|X_n - X| \geq \epsilon/2) + P(|X_n - Y| \geq \epsilon/2).$$

Since $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{P} Y$, as $n \rightarrow \infty$, both terms on the right go to 0. $P(|X_n - X| \geq \epsilon/2) \rightarrow 0$. $P(|X_n - Y| \geq \epsilon/2) \rightarrow 0$. Since the left side $P(|X - Y| \geq \epsilon)$ does not depend on n , it must be that for any $\epsilon > 0$, $P(|X - Y| \geq \epsilon) = 0$. Now consider the event $\{X \neq Y\} = \bigcup_{k=1}^{\infty} \{|X - Y| \geq 1/k\}$. Since $P(|X - Y| \geq 1/k) = 0$ for all $k \geq 1$, and the union of countably many sets of probability zero also has probability zero (by σ -subadditivity or direct property of measures):

$$P(X \neq Y) = P\left(\bigcup_{k=1}^{\infty} \{|X - Y| \geq 1/k\}\right) \leq \sum_{k=1}^{\infty} P(|X - Y| \geq 1/k) = \sum_{k=1}^{\infty} 0 = 0.$$

Thus, $P(X \neq Y) = 0$, which means $P(X = Y) = 1$.

- (iv) From part (ii) we conclude that $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{P} Y$ and from part (iii) we directly get $P(X = Y) = 1$.

Exercise 4 (convergence of normal random variables). Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of RVRVs such that $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$ for each $n \in \mathbb{N}$.

- (i) Show that if $\mu_n \rightarrow \mu \in \mathbb{R}$ and $\sigma_n^2 \rightarrow \sigma^2 > 0$ as $n \rightarrow \infty$, then $X_n \xrightarrow{d} X$, where $X \sim \mathcal{N}(\mu, \sigma^2)$.

[Hint: Use characteristic functions with the uniqueness and continuity properties. The cf of $\mathcal{N}(m, s^2)$ is $\phi(t) = \exp(imt - \frac{1}{2}s^2t^2)$.]

- (ii) Show that if $\mu_n \rightarrow \mu \in \mathbb{R}$ and $\sigma_n^2 \rightarrow 0$ as $n \rightarrow \infty$, then $X_n \xrightarrow{P} \mu$.

[Hint: You can use either Chebyshev's inequality or the fact that convergence in distribution to a constant implies convergence in probability to that constant.]

Solution:

- (i) The characteristic function of $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$ is $\phi_{X_n}(t) = \exp(i\mu_n t - \frac{1}{2}\sigma_n^2 t^2)$. As $n \rightarrow \infty$, we have $\mu_n \rightarrow \mu$ and $\sigma_n^2 \rightarrow \sigma^2$. Therefore, for any fixed $t \in \mathbb{R}$

$$\lim_{n \rightarrow \infty} \phi_{X_n}(t) = \lim_{n \rightarrow \infty} \exp(i\mu_n t - \frac{1}{2}\sigma_n^2 t^2) = \exp(i\mu t - \frac{1}{2}\sigma^2 t^2).$$

This limit is $\phi_X(t)$, the characteristic function of a random variable $X \sim \mathcal{N}(\mu, \sigma^2)$. Since the characteristic functions $\phi_{X_n}(t)$ converge pointwise to $\phi_X(t)$ for all $t \in \mathbb{R}$, and $\phi_X(t)$ is the cf of a $\mathcal{N}(\mu, \sigma^2)$ distribution (which is continuous at $t = 0$), by Lévy's continuity theorem, $X_n \xrightarrow{d} X$.

- (ii) We want to show that for any $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|X_n - \mu| \geq \epsilon) = 0$. Consider $Y_n = X_n - \mu_n$. Then $Y_n \sim \mathcal{N}(0, \sigma_n^2)$. We are interested in $P(|X_n - \mu| \geq \epsilon)$. We can write $X_n - \mu = (X_n - \mu_n) + (\mu_n - \mu)$. Using Chebyshev's inequality for X_n with $\mathbb{E}[X_n] = \mu_n$, $\text{var}[X_n] = \sigma_n^2$ we get

$$P(|X_n - \mu_n| \geq \epsilon) \leq \frac{\text{var}[X_n]}{\epsilon^2} = \frac{\sigma_n^2}{\epsilon^2}.$$

Since $\sigma_n^2 \rightarrow 0$, we have $\lim_{n \rightarrow \infty} P(|X_n - \mu_n| \geq \epsilon) = 0$. This means $X_n - \mu_n \xrightarrow{P} 0$. We know that $\mu_n \rightarrow \mu$, which means $\mu_n - \mu \rightarrow 0$ (just convergence of real numbers, which implies convergence in probability viewed as random variables $\mu_n - \mu \xrightarrow{P} 0$). Since $X_n - \mu = (X_n - \mu_n) + (\mu_n - \mu)$, and $X_n - \mu_n \xrightarrow{P} 0$ and $\mu_n - \mu \xrightarrow{P} 0$, their sum also converges in probability to $0 + 0 = 0$ (according to Exercise 3(i)). Thus, $X_n - \mu \xrightarrow{P} 0$, which is equivalent to $X_n \xrightarrow{P} \mu$.

Alternatively, from part (i), if $\sigma_n^2 \rightarrow 0$, then $X_n \xrightarrow{d} \mathcal{N}(\mu, 0)$. A normal distribution with zero variance is a "point mass" at its mean, i.e., a constant μ (or "deterministic RV"). Convergence in distribution to a constant implies convergence in probability to that constant. Therefore, $X_n \xrightarrow{P} \mu$.

Exercise 5 (bonus: basic properties of ϕ_X). Let X and $(X_i)_{i=1}^n$ be RVRs. Verify the following properties of the characteristic function.

- (i) $|\phi_X(t)| \leq 1$ for all $t \in \mathbb{R}$ and $\phi_X(0) = 1$.
- (ii) $\phi_{aX+b}(t) = e^{ibt} \phi_X(at)$ for all $a \in \mathbb{R} \setminus \{0\}, b \in \mathbb{R}$.
- (iii) We have that $P_X = P_{-X}$ if and only if ϕ_X is real-valued.
- (iv) If $X_1, \dots, X_n$ are independent, then

$$\phi_{X_1 + \dots + X_n}(t) = \phi_{X_1}(t) \cdots \phi_{X_n}(t).$$

Solution:

- (i) We compute

$$\begin{aligned} |\phi_X(t)| &= |\mathbb{E}[e^{itX}]| \\ &\leq \mathbb{E}[|e^{itX}|] \\ &= \mathbb{E}[|\cos(tX) + i \sin(tX)|] \\ &= \mathbb{E}[\sqrt{\cos^2(tX) + \sin^2(tX)}] \\ &= \mathbb{E}[1] = 1. \end{aligned}$$

Further, $\phi_X(0) = \mathbb{E}[e^{i \cdot 0 \cdot X}] = \mathbb{E}[e^0] = \mathbb{E}[1] = 1$.

- (ii) We compute

$$\phi_{aX+b}(t) = \mathbb{E}[e^{it(aX+b)}] = \mathbb{E}[e^{itaX+itb}] = \mathbb{E}[e^{itaX} e^{itb}] \stackrel{(*)}{=} e^{itb} \mathbb{E}[e^{i(at)X}] = e^{itb} \phi_X(at).$$

In (*), e^{itb} is a constant with respect to the expectation.

- (iii) $P_X = P_{-X}$ means that X and $-X$ have the same distribution. This implies they have the same characteristic function, i.e., $\phi_X(t) = \phi_{-X}(t)$. We know from (ii) that $\phi_{-X}(t) = \phi_{(-1)X+0}(t) = e^{i \cdot 0 \cdot t} \phi_X(-1 \cdot t) = \phi_X(-t)$. Also,

$$\phi_X(-t) = \mathbb{E}[e^{i(-t)X}] = \mathbb{E}[\cos(-tX) + i \sin(-tX)] = \mathbb{E}[\cos(tX) - i \sin(tX)] = \overline{\phi_X(t)},$$

where the bar denotes complex conjugation. So, $P_X = P_{-X}$ if and only if $\phi_X(t) = \overline{\phi_X(t)}$ for all $t \in \mathbb{R}$. A complex number z satisfies $z = \bar{z}$ if and only if its imaginary part is zero, i.e., z is real-valued. Thus, $P_X = P_{-X}$ if and only if $\phi_X(t)$ is real-valued for all $t \in \mathbb{R}$.

- (iv) Let $S_n = X_1 + \cdots + X_n$. Then

$$\phi_{S_n}(t) = \mathbb{E}[e^{itS_n}] = \mathbb{E}[e^{it(X_1 + \cdots + X_n)}] = \mathbb{E}[e^{itX_1} e^{itX_2} \cdots e^{itX_n}].$$

Since $X_1, \dots, X_n$ are independent, the random variables $e^{itX_1}, \dots, e^{itX_n}$ are also independent (as functions of independent RVs). Therefore, the expectation of their product is the product of their expectations and we get

$$\phi_{S_n}(t) = \mathbb{E}[e^{itX_1} \cdots e^{itX_n}] = \mathbb{E}[e^{itX_1}] \cdots \mathbb{E}[e^{itX_n}] = \phi_{X_1}(t) \cdots \phi_{X_n}(t).$$